

Complete FAI Governance Operational Test Library

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Date: May 15, 2026

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Series: D5.07 — Phase D5, Note #642

Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in Inter-Self Coordination via Shared Substrate / Full Aspect Integration (Li, April 2026), the third paper in the CKS theory series following Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems (Li, April 2026) and The Instinct/Reasoning Separation Outside the Model (Li, April 2026).

Abstract

This note is the consolidated reference for the complete FAI governance operational test library. The library contains 55 binary pass/fail tests organized into six categories: the 20-test D2.24 baseline suite, and 35 tests from five expansion categories introduced in D5.02 through D5.06. Together the 55 tests serve two functions: compliance verification (confirming that FAI events and relationships meet Paper 3's governance commitments) and prior-art demonstration (establishing that the prior-art corpus covers FAI governance requirements in sufficient operational detail to define binary pass/fail tests). Tests 1–20 are listed here by number and name only; their full text is in D2.24. Tests 21–55 are presented in the compact reference format with binary question and pass/fail interpretation for each test. Usage guidance for three governance contexts closes the note.

1. The Library's Two Functions

The 55-test library serves governance practitioners and the prior-art record in distinct but complementary ways.

Function 1 — Compliance verification. Governance practitioners use the library to confirm that a given FAI event, FAI relationship, or inter-Self governance arrangement satisfies the commitments Paper 3 introduces. Tests 1–20 constitute the minimum compliance checklist. Tests 21–55 extend beyond minimum compliance to assess quality, detect anti-patterns, and probe properties that distinguish governance-motivated inter-Self coordination from superficially similar arrangements that lack architectural depth.

Function 2 — Prior-art demonstration. The library demonstrates that the prior-art corpus reaches governance requirements at operational specificity. Each test derives from a specific note in the D-series derivation sequence, and each test's binary form requires that the relevant commitment be understood precisely enough to distinguish a passing from a failing case. Any adversarial claim that Paper 3's governance requirements are vague or incomplete must address the 55-test library.

The compact reference format — binary question, pass condition, fail condition — is chosen to make the library useful to practitioners directly, without requiring them to first read the establishing notes. Practitioners who want the full derivation behind any test can find it in the establishing note cited in brackets.

2. Tests 1–20: D2.24 Baseline Operational Test Suite

The 20 baseline tests were established in D2.24. They cover the core lifecycle and structural governance requirements for FAI events and inter-Self relationships. The full text of each test — including the binary question and pass/fail interpretation — is in D2.24. This section lists the tests by number and name for reference; readers performing compliance verification should consult D2.24 for the complete test text before running these tests.

The 20 tests span ten governance domains, with two tests per domain:

Construction governance (Tests 1–2)

- Test 1: Shared Substrate Construction Authority
- Test 2: Shared Substrate Construction Record

Dissolution governance (Tests 3–4)

- Test 3: Dissolution Completion Verification
- Test 4: Dissolution Persistence Policy

Configuration completeness (Tests 5–6)

- Test 5: Configuration as Substrate Content
- Test 6: Configuration Completeness Check

Exchange bounding (Tests 7–8)

- Test 7: Exchange Bounding to DNA and Action Layers
- Test 8: Exchange Bounding Provenance

Conflict registry integrity (Tests 9–10)

- Test 9: Conflict Registry Integrity
- Test 10: Conflict Classification Record

Escalation routing (Tests 11–12)

- Test 11: Intra-Tier Escalation Routing
- Test 12: Cross-Perimeter Escalation Routing

Evolution feed mechanics (Tests 13–14)

- Test 13: Evolution Feed Hand-Off Completeness
- Test 14: Evolution Feed Layer Routing

Home perimeter integrity (Tests 15–16)

- Test 15: Home Perimeter Integrity Preservation

- Test 16: Home Perimeter Independence

Path retraceability (Tests 17–18)

- Test 17: Path Retraceability Within Shared Substrate
- Test 18: Path Retraceability Across Perimeters

Governance record completeness (Tests 19–20)

- Test 19: Governance Record Completeness
- Test 20: Governance Record Accessibility

Full test text for Tests 1–20: see D2.24.

3. Tests 21–55: Expansion Categories

Tests 21–55 were introduced in D5.02 through D5.06. Each test is presented in the compact reference format: test number, name, establishing note in brackets, the binary question, the pass condition, and the fail condition. Tests can be run independently; no test in this section requires running another test first.

Category A: Composition Pair Tests (Tests 21–30)

Establishing note: D5.02. These ten tests probe whether a given FAI event or inter-Self governance arrangement correctly instantiates Paper 3's commitments at the points where two or more architectural commitments must hold jointly. A composition pair test fails when one commitment is satisfied but a second commitment — which must hold simultaneously — is not.

[TEST 21] Shared Substrate × Paper 1 Hybrid Commitment (D5.02)

Q: Does the shared substrate maintain the substrate-LLM governance boundary throughout the FAI event — specifically, is AI operating only as mediator over substrate content, with no AI operation bypassing the substrate to exchange information directly between Selves?

PASS: The shared substrate carries the substrate-LLM governance boundary as a structural property: every AI operation during the FAI event is a mediated operation over substrate content, and no inter-Self exchange occurs outside the shared substrate's governance perimeter. Paper 1's hybrid commitment holds within the shared substrate's scope.

FAIL: AI operates outside the shared substrate during the FAI event — for example, LLM-to-LLM message passing that bypasses the shared substrate, or direct model-output exchange that occurs before results are written to substrate. Indicates that the shared substrate is being used as a coordination record while a parallel opaque exchange channel is active, which violates the hybrid commitment at the inter-Self scope.

[TEST 22] FAI × Instinct/Reasoning Separation (D5.02)

Q: Does the FAI event exchange only DNA-layer and action-layer substrate content — and specifically, does no LLM-weight content, instinct-layer parameter content, or instinct-layer inference content cross any Self's perimeter during the event?

PASS: The exchange is bounded to DNA-layer and action-layer content as Paper 3 requires. Each participating Self's instinct layer remains entirely within its own perimeter throughout; the FAI event does not modify, sample, or transmit any content from any Self's instinct layer.

FAIL: The exchange reaches into instinct-layer content in any participating Self — whether by transmitting inference outputs that are not first written to substrate, exchanging model weights, or

including instinct-layer parameters in the shared substrate. Indicates that the instinct/reasoning separation established in Paper 2 is not preserved at the inter-Self boundary, which is a foundational violation of the Paper 3 architecture.

[TEST 23] Three-Tier Conflict Handling × FAI Exchange Bounding (D5.02)

Q: When a conflict surfaces within the shared substrate during an FAI event, is the conflict handled within the three-tier framework — preserve, resolve via orchestration, or escalate to humans — without any handling action that moves conflict resolution outside the shared substrate?

PASS: Every conflict that surfaces during the FAI event receives a three-tier classification and is handled within the shared substrate accordingly. No conflict is resolved through a side channel (private negotiation between human participants, out-of-band AI inference, or governance decisions not written to the shared substrate).

FAIL: One or more conflicts during the FAI event are handled outside the shared substrate — resolved privately, suppressed without record, or disposed of through a mechanism not visible in the governance record. Indicates that conflict handling and exchange bounding are not held jointly: the shared substrate carries exchange content but not governance content, which breaks the architecture.

[TEST 24] Evolution Feed × Home-Perimeter Independence (D5.02)

Q: After the FAI event's evolution feed hand-off, does each participating Self's home evolution mechanism operate on the handed-off content independently under its own governance — specifically, without any participating Self having authority over how another Self's home mechanisms ingest or apply the content?

PASS: Each Self's home evolution mechanisms receive the evolution feed outputs and process them under that Self's own governance configuration. No inter-Self governance authority carries over past the dissolution boundary; each Self's evolution decisions are entirely within its own home-perimeter authority.

FAIL: One Self's governance authority persists past dissolution to constrain or direct another Self's ingestion of evolution feed content. Indicates that the home-perimeter independence property, which must hold jointly with the evolution feed commitment, has been broken — the FAI event has become an ongoing governance authority relationship rather than a bounded coordination event.

[TEST 25] Configuration × Recursive Applicability (D5.02)

Q: Are all configurable dimensions of the FAI event — including the configuration of how configuration itself is governed — recorded as substrate content within the shared substrate, with no configurable dimension living outside human-governed substrate?

PASS: The full set of configurable dimensions is substrate content: which aspects each Self contributes, merge depth, conflict-handling defaults, persistence policy, and the governance rules for modifying any of these. Configuration of configuration (meta-governance) is also substrate content under human authority. The recursion bottoms at human-authored authority per Paper 1.

FAIL: One or more configurable dimensions are not substrate content — for example, a default merge depth that is a system property not written to substrate, or a conflict-handling rule that is an implementation choice outside human authority. Indicates that the configuration completeness and recursive applicability requirements are not held jointly, leaving a portion of governance outside the shared substrate's perimeter.

[TEST 26] Shared Substrate × Linear-Cost Composition (D5.02)

Q: Does adding a third (or nth) participating Self to the FAI event require only that Self's governance configuration and aspect contributions to be added to the shared substrate — specifically, without

requiring architectural redesign, non-linear increases in coordination overhead, or new coordination layers not already present in the two-Self case?

PASS: The shared substrate scales to n participants by composition: each new Self adds its governance configuration and contributed aspects, and the existing coordination architecture handles the additional participant without structural change. Cost grows linearly with participants, not super-linearly.

FAIL: Adding a participant requires architectural intervention, introduces coordination overhead that is non-linear in the number of participants, or requires a new coordination layer not present in the two-Self case. Indicates that the linear-cost composition property from Paper 1 is not inherited at the inter-Self scope, breaking the composition pair.

[TEST 27] FAI × N-Ary Cardinality (D5.02)

Q: Does the governance configuration of the FAI event explicitly specify cardinality — the set of participating Selves and their respective contributed aspects — as substrate content, rather than treating cardinality as an implicit or external property?

PASS: The FAI event's governance configuration contains an explicit, human-authored specification of which Selves participate and which aspects each contributes. Cardinality is a first-class governance property, adjustable within human authority, and its specification is retained in the governance record.

FAIL: Cardinality is determined implicitly — by which Selves happen to respond to an initiation signal, by a platform default not written to substrate, or by a configuration outside human authority. Indicates that the n -ary cardinality property is not instantiated as a governed substrate property, leaving participation itself ungoverned.

[TEST 28] Conflict Preservation × Path Retraceability (D5.02)

Q: For every conflict preserved in the shared substrate during the FAI event, does the governance record contain a complete path from the originating content contributions through the conflict's classification and preservation decision, such that a reviewer can reconstruct why the conflict was classified and preserved rather than resolved?

PASS: Every preserved conflict has a traceable path: the content contributions that generated the conflict are identifiable, the classification decision is recorded with its governing orchestration rule, and the preservation record links the conflict to its source contributions. No preserved conflict appears in the governance record without a traceable origin.

FAIL: One or more preserved conflicts in the governance record lack a traceable path — their origin in contributing aspects is not recoverable, or their classification decision references an orchestration rule not recorded in the shared substrate. Indicates that conflict preservation and path retraceability are not held jointly: conflicts are recorded but not governably traceable.

[TEST 29] Evolution Feed × Four-Locus Routing (D5.02)

Q: Does the evolution feed hand-off at FAI dissolution correctly route outputs to each of the four evolution loci — DNA evolution, action-feedback evolution, home-perimeter governance updates, and configuration evolution — according to the governance configuration, with no outputs deposited without a routing destination?

PASS: The evolution feed hand-off is fully routed: each output item has a designated locus (one of the four), the routing follows the governance configuration for this event, and the hand-off record contains a complete mapping of output items to loci. No output is unrouted or deposited into a generic store without locus specification.

FAIL: One or more evolution feed outputs lack routing specification — deposited into a home substrate without locus designation, routed to a default locus not specified in the governance configuration, or lost at dissolution without entering the hand-off record. Indicates that the evolution feed and four-locus routing commitments are not held jointly: the feed mechanism exists but routing governance is incomplete.

[TEST 30] Population Scope × Joint Authority (D5.02)

Q: When an FAI event involves Selves operating under distinct governance authority structures (different organizations, different governance perimeters), is the joint authority configuration — which decisions require agreement across perimeters and which are governed within each Self's home perimeter — explicitly specified as substrate content?

PASS: The joint authority configuration is explicit substrate content: the FAI event's governance record identifies which decisions are joint (requiring cross-perimeter agreement) and which are home-perimeter decisions (within each Self's own authority). No decision that crosses perimeters is made without an explicit joint-authority configuration governing it.

FAIL: Cross-perimeter decisions are made without explicit joint-authority specification — governed implicitly, by default, or by a configuration external to the shared substrate. Indicates that the population-scope joint authority commitment is not instantiated, leaving inter-organizational governance undefined within the architecture.

Category B: Anti-Pattern Detection Tests (Tests 31–40)

Establishing note: D5.03. These ten tests detect the anti-patterns Paper 3 identifies as failure modes: configurations that superficially resemble FAI governance while violating its architectural commitments. Each test isolates a specific failure mode; a failure result indicates the anti-pattern is present, not merely that a property is absent.

[TEST 31] Opaque Inter-Self Communication Detection (D5.03)

Q: Is all substantive content exchanged between participating Selves during or in connection with an FAI event written to the shared substrate as human-governed substrate content — with no substantive exchange conducted through message-passing, direct model-to-model calls, or shared-memory access patterns whose content is not first-class substrate content?

PASS: No opaque exchange channel is operating alongside or in lieu of the shared substrate. All substantive content — not just final outputs but reasoning, intermediate states, and conflict-relevant content — is substrate content under human governance.

FAIL: Substantive content is exchanged through a channel that bypasses the shared substrate: LLM-to-LLM message passing, tool-call outputs not written to substrate, shared-memory patterns outside the substrate perimeter, or post-event communication that modifies each Self's home substrate without entering the governance record. This is the primary architectural anti-pattern Paper 3 is designed to replace.

[TEST 32] Shallow Merge / Federation-Only Detection (D5.03)

Q: Does the FAI event's merge operation carry provenance to the depth specified in the governance configuration — including cell-level content, DNA-layer logic, and conflict structure from contributing aspects — rather than merging only surface outputs or federated summaries that lack the architectural depth FAI requires?

PASS: The merge is substantive at the depth the governance configuration specifies. Contributing aspects surface their constituent cells, DNA layer, and action layer within the shared substrate; the

merge operates on this content, not on pre-summarized or pre-federated outputs.

FAIL: The merge operates only on surface outputs — summaries, conclusions, or federated results — without carrying the contributing aspects' cell structure, DNA-layer logic, or conflict history into the shared substrate. This is the shallow merge / federation-only anti-pattern: the architecture uses FAI vocabulary but does not perform the aspect-level exchange Paper 3 commits to.

[TEST 33] Silent Conflict Resolution Detection (D5.03)

Q: Is every conflict that surfaces during the FAI event — including conflicts between contributing aspects, conflicts between aspect content and orchestration rules, and conflicts arising from cross-perimeter assumption divergence — classified and handled within the three-tier framework, with no conflict suppressed, discarded, or resolved without a record?

PASS: The conflict registry contains a record for every conflict that surfaces. No conflict is silently resolved through AI inference, system default, or human negotiation outside the governance record. Every resolution is traceable to a governing orchestration rule.

FAIL: One or more conflicts are resolved without entering the conflict registry — silently unified by the mediating AI, resolved through default behavior not specified in governance configuration, or negotiated between participants outside the shared substrate. This is the silent conflict resolution anti-pattern: it produces a shared substrate that appears conflict-free while suppressing real governance information.

[TEST 34] Ungoverned Evolution Feed Detection (D5.03)

Q: Is every item in the FAI event's evolution feed output — every piece of content that will enter any participating Self's home evolution mechanisms — covered by an explicit routing specification in the governance configuration, with no item entering home evolution mechanisms through an ungoverned channel?

PASS: The evolution feed is fully governed: every output item has a routing specification, every routing specification is substrate content under human authority, and no output enters a home evolution mechanism through a channel outside the governance record.

FAIL: One or more evolution feed items enter home evolution mechanisms through an ungoverned channel — deposited through a background sync not specified in governance configuration, absorbed through a direct AI operation outside the substrate, or carried over implicitly without a routing record. This is the ungoverned evolution feed anti-pattern: content shapes a Self's home evolution without governance visibility.

[TEST 35] Fixed Configuration Detection (D5.03)

Q: Can the FAI event's governance configuration be modified by humans with appropriate authority — before, during (where governance rules permit mid-event modification), or between events — without requiring a platform change, vendor intervention, or architectural modification outside the substrate?

PASS: The governance configuration is substrate content under human authority: a human with appropriate access can read it, modify it, and have the modification take effect within the substrate's governance perimeter. Modification does not require external intervention.

FAIL: One or more dimensions of the governance configuration cannot be modified by humans with appropriate authority — locked by platform defaults, vendor policy, or architectural constraints not subject to the human authority commitment. This is the fixed configuration anti-pattern: governance vocabulary is present but the configuration is not actually substrate content under human authority.

[TEST 36] Meta-Governance Escape Detection (D5.03)

Q: Does the governance configuration of the FAI event cover not only first-order governance (how content is exchanged and conflicts handled) but also meta-governance (how the governance configuration itself is authorized, modified, and bounded) — with no meta-governance dimension living outside the substrate?

PASS: Meta-governance is fully contained within the substrate: the rules governing who can modify the governance configuration are themselves substrate content under human authority. The recursion bottoms at human-authored authority per Paper 1, with no meta-governance dimension escaping substrate governance.

FAIL: Meta-governance escapes the substrate — for example, governance configuration can be modified by a platform administrator outside the shared substrate's governance perimeter, or the rules for modifying governance rules are not themselves substrate content. This is the meta-governance escape anti-pattern: first-order governance is in the substrate, but the governance of governance is not.

[TEST 37] Emergent Coordination Detection (D5.03)

Q: Is the coordination that occurs during the FAI event fully explainable by the governance configuration and the substrate content — specifically, is there no coordination outcome (merge decision, conflict classification, evolution routing) that is produced by AI inference operating outside the governance configuration without a substrate record of the governing rule applied?

PASS: Every coordination outcome during the FAI event has a traceable governance basis: the orchestration rule that governs it is substrate content, the AI mediator applied that rule, and the application is recorded in the governance record. No outcome is produced by AI inference operating outside the governance framework.

FAIL: One or more coordination outcomes cannot be traced to a governing orchestration rule in the substrate — produced by AI inference from context, by emergent behavior across multiple mediator operations, or by a default not recorded in the governance configuration. This is the emergent coordination anti-pattern: coordination happens, but it is not governed.

[TEST 38] LLM-Weight Exchange Detection (D5.03)

Q: Does the FAI event avoid any exchange of LLM-weight content, instinct-layer parameters, or fine-tuning data between participating Selves — including indirectly, through exchanges of content that encodes model-internal representations not derived from substrate content?

PASS: No content deriving from LLM weights or instinct-layer parameters crosses any Self's perimeter during the FAI event. The exchange is bounded to substrate-derived content (DNA-layer and action-layer content as specified in Paper 3), and the exchange bounding record in the governance configuration confirms this.

FAIL: LLM-weight content, instinct-layer parameters, or fine-tuning data crosses a Self's perimeter — directly or through encoded representations of model-internal states not derived from substrate content. Indicates that the instinct/reasoning separation from Paper 2 has been violated at the inter-Self boundary.

[TEST 39] Instinct Boundary Violation Detection (D5.03)

Q: Does the FAI event's evolution feed avoid routing any output to a participating Self's instinct layer — specifically, does no FAI output enter a Self's LLM weights, instinct-layer training data, or any other mechanism through which FAI content could directly modify a Self's instinct-layer inference behavior?

PASS: The evolution feed routing configuration directs all outputs to DNA-layer and action-layer evolution mechanisms, home-perimeter governance updates, or configuration evolution. No output is routed to any Self's instinct layer. The routing record in the governance configuration confirms

instinct-layer exclusion.

FAIL: One or more evolution feed outputs are routed to a Self's instinct layer — entering fine-tuning data, prompt-construction mechanisms that modify instinct-layer behavior, or any pathway through which FAI content directly shapes LLM inference. This anti-pattern breaks the instinct/reasoning separation at the evolution feed hand-off point.

[TEST 40] Opaque Persistence Policy Detection (D5.03)

Q: Is the persistence policy for the FAI event — what is retained after dissolution, in what form, under whose governance authority, and for how long — explicitly specified as substrate content before dissolution occurs, with no persistence decision made implicitly, by platform default, or outside human authority?

PASS: The persistence policy is explicit substrate content: the governance configuration specifies exactly what is retained, under whose governance authority, and for what duration. Dissolution executes the specified policy.

FAIL: The persistence policy is not explicit substrate content — governed by platform default, determined after dissolution by whichever participant acted first, or not specified at all. This is the opaque persistence policy anti-pattern: what survives the FAI event, and whose governance authority covers it, is determined outside the architecture.

Category C: Governance Quality Tests (Tests 41–45)

Establishing note: D5.04. These five tests assess the quality of governance implementation beyond minimum compliance. A minimum-compliant FAI event passes Tests 1–20; a high-quality FAI governance implementation passes Tests 41–45 as well. These tests do not probe for anti-patterns; they distinguish thin compliance from architecturally robust governance.

[TEST 41] Configuration Depth Assessment (D5.04)

Q: Does the governance configuration of the FAI event specify all seventeen configurable dimensions identified across Paper 3's core claims — covering participating aspects, merge depth, conflict-handling defaults for each tier, persistence policy, evolution routing for each locus, joint-authority scope, and meta-governance rules — rather than specifying only the dimensions required to run the event?

PASS: The governance configuration is complete across all seventeen configurable dimensions. Each dimension is explicitly specified, not defaulted implicitly. A reviewer reading the governance configuration can reconstruct the full governance logic of the FAI event without consulting any source outside the shared substrate.

FAIL: One or more configurable dimensions are not specified — relying on system defaults, implicit behavior, or configuration external to the substrate. A thin configuration that specifies only the dimensions required to prevent error is compliant with the minimum baseline but does not meet the depth standard governance quality requires.

[TEST 42] Cross-Perimeter Escalation Pathway Quality (D5.04)

Q: For each class of conflict that might require cross-perimeter escalation — conflicts that cannot be resolved within the shared substrate's orchestration rules and require human authority from more than one participating Self's governance structure — does the governance configuration specify a named escalation pathway, including who holds escalation authority and by what mechanism they are notified?

PASS: The governance configuration contains a named escalation pathway for every conflict class that might require cross-perimeter human authority. Each pathway identifies the humans who hold authority, the notification mechanism, and the governance rule for how the escalation is recorded in the shared substrate.

FAIL: Cross-perimeter escalation is not pre-specified — to be determined when a conflict arises, handled informally, or absent from the governance configuration. A minimum-compliant implementation may have escalation routing (Test 12) without having pre-specified pathway quality; the absence of named pathways means governance depends on improvised coordination at the moment escalation is needed.

[TEST 43] Persistence Policy Specificity (D5.04)

Q: Does the persistence policy specify not only what is retained after dissolution but also the governance chain of custody for retained content — specifically, which governance authority holds each retained content element, and by what mechanism a party seeking to inspect, modify, or override retained content can exercise those rights?

PASS: The persistence policy includes chain-of-custody specification: for each retained content element, the governing authority is named, and the mechanism for exercising inspect/modify/override rights is specified in the governance configuration. Retained content does not exist in a governance limbo after dissolution.

FAIL: The persistence policy specifies what is retained but not governance chain of custody — retained content exists but its governance authority after dissolution is ambiguous or unspecified. This is a quality gap rather than a compliance failure: retained content that lacks chain-of-custody specification may be technically accessible but practically ungoverned.

[TEST 44] Governance Record Auditability (D5.04)

Q: Can a reviewer who was not present during the FAI event reconstruct the complete governance history of the event — every governance decision, every orchestration rule applied, every conflict classified, and every evolution output routed — from the governance record alone, without access to external context, participant memory, or system logs outside the shared substrate?

PASS: The governance record is self-contained and complete: every governance action during the event is recorded with its governing rule, its inputs, and its outputs. A reviewer who had no prior knowledge of the event can reconstruct the full governance history from the record.

FAIL: The governance record requires supplementation — external context to interpret, participant testimony to complete, or system logs to verify. A record that passes Tests 19–20 may still fail Test 44 if the record's completeness depends on context not contained within the shared substrate. High-quality governance produces records that are self-contained by design, not by circumstance.

[TEST 45] Human Authority Accessibility Under FAI (D5.04)

Q: During an active FAI event, can a human with appropriate authority inspect any shared substrate content, modify any orchestration rule, or halt the event — specifically without requiring a platform intervention, vendor action, or pause-and-restart sequence that introduces delay beyond what the governance configuration specifies as acceptable?

PASS: Human authority is architecturally accessible during an active FAI event: the inspect, modify, and override rights from Paper 1's human-governed commitment are exercisable in real time, within the shared substrate's governance perimeter, without requiring any action outside the architecture.

FAIL: Human authority during an active FAI event is constrained by platform mechanics, vendor policy, or architectural bottlenecks not specified in the governance configuration — requiring external intervention, platform-level pauses, or delays not governed within the shared substrate. This is a

quality gap in the human-governed property at inter-Self scope.

Category D: Evolution Learning Tests (Tests 46–50)

Establishing note: D5.05. These five tests assess whether the FAI event's evolution feed is not merely formally complete but architecturally sound — producing outputs that will actually support each Self's home evolution mechanisms in the ways Paper 3's Claim 4 commits to.

[TEST 46] Evolution Feed Completeness (D5.05)

Q: Does the evolution feed hand-off at FAI dissolution include, for each participating Self, all content from the shared substrate that is relevant to that Self's four evolution loci — and specifically, does it include conflict annotations that surface boundary conditions the Self will need to address within its home governance, not only positive exchange content?

PASS: The evolution feed is complete for each participating Self: it includes positive content (aspects, protocols, logic accumulated during the FAI event), conflict annotations (boundaries and assumptions that diverged during the event), and any configuration evolution content that updates this Self's governance logic for future FAI events.

FAIL: The evolution feed is incomplete — including only positive content while omitting conflict annotations, or including only the content from one Self's contribution while omitting content derived from the composition. An incomplete evolution feed means the FAI event's value for each Self's learning is truncated.

[TEST 47] Layer Routing Accuracy (D5.05)

Q: Does the evolution feed routing correctly classify each output item by its target layer — DNA-layer evolution, action-layer (action-feedback) evolution, home-perimeter governance update, or configuration evolution — such that each item reaches the home mechanism equipped to process it, rather than being deposited at a generic substrate layer?

PASS: Every evolution feed item is routed to the correct layer: DNA-layer content to DNA evolution, action-layer feedback content to action-feedback evolution, governance changes to home-perimeter governance, and configuration updates to configuration evolution. The routing record documents the classification basis for each item.

FAIL: One or more items are misrouted — DNA-layer content deposited in action-layer mechanisms, action-feedback content absorbed into DNA evolution, or governance content deposited as generic substrate content without a governance mechanism to process it. Misrouting produces formally non-zero evolution outputs while delivering them to the wrong home mechanism.

[TEST 48] Asymmetric Ingestion Governance (D5.05)

Q: Does the governance configuration accommodate asymmetric ingestion — where participating Selves ingest different subsets of the evolution feed outputs based on their respective governance configurations — without treating equal ingestion as a default requirement?

PASS: The governance configuration explicitly specifies each Self's ingestion scope: which evolution feed items each Self ingests, which it receives but does not ingest (retaining for future governance decisions), and which are outside its scope entirely. Asymmetry is governed, not accidental.

FAIL: The governance configuration treats ingestion as symmetric by default — assuming each Self ingests the same evolution content — or does not specify ingestion scope at all. Ungoverned symmetric ingestion means that Selves with narrower governance mandates may be absorbing content that their home governance is not configured to handle.

[TEST 49] Evolution Feed to Home Substrate Integration (D5.05)

Q: After the evolution feed hand-off, can each participating Self's home governance mechanisms accept the handed-off content — is the content in a form compatible with that Self's home substrate format, governance configuration, and evolution mechanisms, as specified in the FAI event's governance configuration?

PASS: The governance configuration includes a compatibility specification: the format and governance-context requirements each Self's home mechanisms expect, and a confirmation that the evolution feed hand-off is formatted to match. Hand-off content is positioned to be actionable within each Self's home evolution framework.

FAIL: The evolution feed hand-off deposits content in a format or governance context incompatible with one or more receiving Selves' home mechanisms — requiring post-dissolution reformatting, governance re-specification, or human intervention to make the content usable. Compatibility failure is a quality gap that Paper 3's Claim 4 requires the governance configuration to prevent.

[TEST 50] FAI Dissolution Hand-Off Integrity (D5.05)

Q: Does the dissolution event complete the evolution feed hand-off atomically — specifically, is there no state in which the shared substrate has partially dissolved (some content removed) while evolution feed outputs have only partially entered receiving Selves' home substrates?

PASS: The dissolution governance configuration specifies a hand-off integrity protocol: dissolution and hand-off are coordinated so that the shared substrate does not release content until receiving Selves have confirmed receipt, or the governance record documents hand-off status at dissolution time in a form that allows resumption if a receiving Self does not confirm receipt.

FAIL: Dissolution occurs independently of hand-off confirmation — the shared substrate dissolves on a timer or trigger not coordinated with receiving Selves' confirmation, with no hand-off integrity protocol specified. Partial hand-off at dissolution means evolution content is lost, and the governance record may not surface the loss because the dissolution event is recorded as complete.

Category E: Population-Scope Tests, Calibrated-Humility Register (Tests 51–55)

Establishing note: D5.06. These five tests address Paper 3's Claim 6 territory — population-scale governance across many participating Selves — and apply the calibrated-humility register appropriate to extension-claim content. Passing a population-scope test confirms that the architectural commitments hold at population scale; it does not confirm any empirical claim about what population-scale FAI will achieve in practice.

[TEST 51] Population-Level Joint Authority (D5.06)

Q: When FAI events accumulate across a network of participating Selves under distinct governance authority structures, is joint authority across population-level governance perimeters specified as substrate content — specifically, is there a population-level governance configuration that covers which decisions operate under joint authority and which remain within each Self's home perimeter, and is this configuration substrate content under human authority?

PASS: Population-level joint authority is explicit substrate content: the governance configuration covers joint-authority scope at population scale, it is under human authority per Paper 1, and it is accessible to governance review. The population-level governance configuration is not an emergent property of accumulated FAI events but an authored governance artifact.

FAIL: Population-level joint authority is implicit — determined by accumulated FAI event configurations, by platform defaults operating at population scale, or by inter-organizational agreements not written to substrate. This is the population-scope version of the meta-governance

escape anti-pattern: the governance architecture commits to substrate-mediated coordination at the individual FAI event level while leaving population-scale governance ungoverned.

[TEST 52] Accumulated FAI Event Record (D5.06)

Q: Does the governance infrastructure supporting population-scale FAI maintain a record of accumulated FAI events in a form that is accessible to population-level governance review — specifically, in a form that enables governance practitioners to assess what collective evolution dynamics are operating, without relying on AI inference over opaque event logs?

PASS: The accumulated FAI event record is substrate content at population scope: events are recorded in a human-accessible form, the record is under human authority (inspect, modify, override rights are available to population-level governance authorities), and the record is sufficient for governance practitioners to assess which patterns are propagating, which Selves are participating, and which conflict boundaries are recurring at population scale.

FAIL: The accumulated FAI event record is an opaque log — technically complete but requiring AI inference, specialized tooling outside human governance authority, or platform-mediated access to interpret. An opaque accumulation record satisfies a literal compliance reading while breaking the governance-motivated commitment: humans can govern collective dynamics only if the accumulated record is legible to them directly.

[TEST 53] Substrate-Mediator Ladder Rung Positioning (D5.06)

Q: Does the governance configuration of FAI events explicitly identify which rung of the substrate-mediator ladder is operative — distinguishing Cell scope (rungs 1–2), Self scope (rungs 3–4), inter-Self scope (rung 5), and population scope (rung 6) — so that governance practitioners applying these tests know which rung's commitments apply?

PASS: The governance configuration includes a rung-positioning statement: it identifies the operative scope and confirms which commitments apply at that scope. Practitioners running this test library against a given FAI event or governance arrangement can confirm which categories of tests are applicable.

FAIL: No rung-positioning statement exists. Without explicit scope identification, governance practitioners may apply tests from inapplicable rungs or may miss tests that are applicable. The ladder-rung framing is not ornamental: it is the mechanism by which the test library remains accurate as scope grows.

[TEST 54] Calibrated-Humility Register Compliance (D5.06)

Q: Do governance documents and communications associated with population-scale FAI operations — including governance configurations, evolution feed records, and governance reports — apply the calibrated-humility register appropriate to extension-claim content: committing to architectural properties while refraining from empirical predictions about what population-scale FAI will achieve in practice?

PASS: Governance documents apply the calibrated-humility register: they state architectural commitments precisely and confidently (joint authority across population-level governance perimeters is a design property; accumulated FAI events produce collective evolution dynamics through substrate composition) while marking possibilities and open questions as empirical or philosophical questions pending validation. No governance document makes a population-scale capability prediction not supported by architectural argument.

FAIL: Governance documents conflate architectural commitments with empirical predictions — asserting that population-scale FAI will converge to broadly-capable AI, that specific coordination outcomes will obtain, or that the architecture guarantees population-scale results beyond what the

architectural commitments themselves warrant. Calibrated-humility register failure overstates the architecture's current commitments in ways that invite reliability risk when empirical validation does not immediately confirm the prediction.

[TEST 55] Population-Scale Configuration Governance (D5.06)

Q: Is the configuration of population-scale dynamics — including governance of which patterns propagate across the network, which Selves can initiate population-level FAI events, and which dimensions of population-level coordination are joint-authority vs. home-perimeter decisions — itself substrate content under joint human authority at population scope, with no population-scale configuration dimension escaping human governance?

PASS: Population-scale configuration is fully under human governance: the rules governing which patterns propagate, which Selves participate in population-level events, and how joint authority operates across population-level perimeters are substrate content under the joint authority of population-level governance authorities. The recursion from Paper 1 and Paper 3 Claim 5 bottoms at population-level human governance as specified.

FAIL: One or more population-scale configuration dimensions escape human governance — governed by platform mechanics, by emergent network behavior, or by individual Self configurations that compound into population-level effects outside any explicit joint-authority governance. This is the population-scope version of the fixed configuration anti-pattern: population-scale behavior is determined by the system's accumulated state rather than by governed configuration.

4. Usage Guidance

The 55-test library is organized for use across three governance contexts. Practitioners should begin with the context that matches their current situation; tests from other contexts can be added as depth is needed.

For first-time FAI participants. Run Tests 1–20 (the D2.24 baseline suite) as the minimum compliance checklist before initiating an FAI event. These tests confirm that the shared substrate construction, dissolution, configuration, exchange bounding, conflict handling, escalation routing, evolution feed, home perimeter, path retraceability, and governance record requirements are all in place. A first-time participant who passes all 20 baseline tests is ready to initiate a governed FAI event. Full test text for Tests 1–20 is in D2.24.

For ongoing governance audit. Run Tests 21–55 periodically — after each FAI event in early deployment, and at regular intervals in mature deployment — to assess governance quality beyond minimum compliance. The composition pair tests (21–30) confirm that commitments are being held jointly, not merely independently. The governance quality tests (41–45) assess configuration depth, escalation pathway quality, persistence specificity, record auditability, and real-time human authority. The evolution learning tests (46–50) confirm that the evolution feed is producing outputs that will actually support each Self's home evolution. The population-scope tests (51–55) are applicable when FAI events have begun to accumulate at scale.

For governance investigation. When a governance issue is suspected — an FAI event produced unexpected outcomes, a conflict was not handled in the expected tier, an evolution feed output did not materialize — run the anti-pattern detection tests (31–40) first. Each test isolates a specific failure mode: Test 31 for opaque exchange channels, Test 32 for shallow merge, Test 33 for silent conflict resolution, Test 34 for ungoverned evolution feeds, Tests 35–36 for configuration and meta-governance escape,

Test 37 for emergent coordination, Tests 38–39 for instinct boundary violations, and Test 40 for opaque persistence policy. A positive result on any anti-pattern detection test identifies the specific failure mode for targeted remediation.

5. Library Summary

Tests	Category	Count	Establishing Note	Primary Function
1–20	D2.24 Baseline	20	D2.24	Minimum compliance
21–30	Composition Pairs	10	D5.02	Joint commitment verification
31–40	Anti-Pattern Detection	10	D5.03	Failure mode detection
41–45	Governance Quality	5	D5.04	Quality beyond compliance
46–50	Evolution Learning	5	D5.05	Evolution feed soundness
51–55	Population Scope	5	D5.06	Population-scale governance
Total		55		

How to Cite This Note

Li, W. (2026). *Complete FAI Governance Operational Test Library*. May 15, 2026. ORCID: 0009-0004-8065-3235. Derivation note D5.07 (#642) in the CKS theory derivation series.